

National Exam of Higher National Diploma 2025 Session

Specialty/option: SWE - NWS - EDM

Paper : Discrete Mathematics

Duration : 4 hours

Credits : 4 / 4 / 10

GENERAL INSTRUCTIONS.

- **ANSWER ALL QUESTIONS**
- You are reminded of the necessity of good English and orderly presentation of your material.
- Where calculations are required, clearly show your working and be chronological in your answer.
- You are authorized to use only non-programmable calculator

SECTION - A: MULTIPLE CHOICE QUESTIONS (20 marks)

ANSWER THIS SECTION BY WRITING ONLY THE LETTER CORRESPONDING TO THE CORRECT ANSWER

1) Let $f(x, y, z) = x^2 \ln yz + 2 \cos xy$. Then $f_x(x, y, z) =$

- A. $2x \ln yz - 2y \sin xy$
- B. $2x \ln yz + 2y \sin xy$
- C. $2 \ln yz - 2x \sin xy$
- D. $2 \ln yz + 2y \sin xy$

2) Let $f: \mathbb{R}^3 \rightarrow \mathbb{R}$ be given by $f(x, y, z) = x^2 + 3xy + yz^3$. $\text{grad } f$ is

- A. $(5x, 3y, 3yz^2)$
- B. $(2x + 3y, 3x + z^3, 3y z^2)$
- C. $(x^2 + 3y, 3x + z^3, 3y z^2)$
- D. $2x + 3y, 3x, z^3 + y z^3$

3) Given that $x^2 y + 5x^3 y^4 = 7$, $\frac{dy}{dx} =$

- A. $\frac{2xy + 15x^2 y^4}{x^2 + 20x^3 y^3}$
- B. $-\frac{2xy + 15x^2 y^4}{x^2 + 20x^3 y^3}$
- C. $\frac{-2xy + 15x^2 y^4}{x^2 + 20x^3 y^3}$
- D. $\frac{2xy + 15x^2 y^4}{-x^2 + 20x^3 y^3}$

4) The value of x for which $\left(\frac{1}{25}\right)^x = 5^{2(x+3)}$ is

$\frac{x}{2} = 2(x+3)$
 $x + 10 = 2x$
 $10 = 2x - x$
 $10 = x$
 $2(x) = 2(x+3)$
 $2x = 2x + 6$
 $2x - 2x = 6$
 $0 = 6$

A. $2/3$

~~B. $2/3$~~

C. $3/2$

D. $-3/2$

5) Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a function defined by $f(x) = x^3 + 5x$. Then the function f is

A. Periodic

~~B. even~~

☒ C. odd

D. Parity

6) Let $f(x)$ be a continuous and strictly decreasing function defined on $[a, b]$, $a < b$. Let $f^{-1}(x) = g(x)$, then which of the following is TRUE?

A. $g'(x) > 0$

B. $g'(x) < 0$

C. $g(x) < 0$

D. $g(x) > 0$.

7) If $P(A) = 0.75$, $P(B) = 0.4$, and $P(A \text{ or } B) = 0.9$, $P(A \text{ and } B)$ is?

A. 0.75

B. 0.4

C. 0.65

D. 0.25

8) When a computer goes down there is a 75% chance that it is due to an overload and a 15% chance that is due to software problem. There is an 85% chance that it is due to an overload or a software problem. The probability that the breakdown is only due to a software problem is

A. 0.15

B. 0.7

C. 0.1

D. 0.25

9) A batch of 25 injection-molded parts contains 5 that have suffered excessive shrinkage. Two parts are selected at random without replacement, the probability that the second part selected is one with excessive shrinkage is

A. $1/5$

B. $1/6$

C. $1/30$

D. $4/5$

10) A computer center has three printers A, B and C, which print at different speeds. Programs are routed to the first available printer. The probability that a program is routed to printers A, B and C are 0.6, 0.3 and 0.1 respectively. Occasionally, a printer will jam and destroy a printout. The probability that printers A, B and C will jam are 0.01, 0.05 and 0.04 respectively. Your program is destroyed when a printer jams. What is the probability that printer A is involved?

A. 0.006

B. 0.24

C. 0.25

D. 0.01

Handwritten notes:

$$f(-x) = -f(x) \Rightarrow f(-x) = f(x)$$
$$-x^3 - 5x = x^3 + 5x$$

11) Given that $(x + 2)$ is a factor of the polynomial $f(x) = x^3 - 2x^2 + kx + 2$. The value of k is

A. 6

B. 1

C. 7

☒ D. -7

$$f(-2) = (-2)^3 - 2(-2)^2 + (-2)k + 2 = 0$$

$$-8 - 8 - 2k + 2 = 0$$

$$-2k = 8 + 8 - 2$$

$$k = \frac{-16 + 2}{-2} = 7$$

$$-16 - 2k = 0$$

$$-2k = 16$$

$$k = -8$$

12) The coefficient of the term in x^2 in the Maclaurin expansion of $\ln(1 - 2x)$ is

A. 4

B. 1

C. 2

D. -2

$$\frac{1}{1-2x} = 1 + 2x + 4x^2 + \dots$$

13) $\lim_{x \rightarrow \infty} \frac{\sqrt{2x^2 + 1}}{3x - 5}$

A. $\frac{2}{3}$

☒ B. $\frac{\sqrt{2}}{3}$

C. ∞

D. 0

14) $\int_0^2 \frac{x}{\sqrt{4 + x^2}} dx$

A. $2\sqrt{2}$

B. $2\sqrt{2} - 2$

C. $2\sqrt{2} - 1$

D. $\frac{1}{2}(\sqrt{2} - 1)$

$$\frac{x^2}{\sqrt{4+x^2}} = \frac{x^2}{x^2} \cdot \frac{1}{\sqrt{4/x^2 + 1}} = \frac{1}{\sqrt{4/x^2 + 1}}$$

$$\frac{x^2}{\sqrt{4+x^2}} = \frac{x^2}{\sqrt{4+x^2}} \cdot \frac{1}{x^2} = \frac{1}{\sqrt{4/x^2 + 1}}$$

$$\frac{x^2}{\sqrt{4+x^2}} = \frac{x^2}{\sqrt{4+x^2}} \cdot \frac{1}{x^2} = \frac{1}{\sqrt{4/x^2 + 1}}$$

$$\frac{1}{\sqrt{4/x^2 + 1}} = \frac{1}{\sqrt{4/x^2 + 1}} \cdot \frac{x}{x} = \frac{x}{\sqrt{4 + x^2}}$$

15) Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be given by $f(x) = \sqrt{1 - 2x}$. The domain of f is

A. $x \leq \frac{1}{2}$

B. $x \geq \frac{1}{2}$

C. $x \leq 2$

D. $x \geq 2$

16) The solution of the equation $|2x - 3| = 3$ is

☒ A. 0

B. 3

C. 0 and 3

D. -3

$$2x - 3 = 3$$

$$2x = 6$$

$$x = 3$$

$$2x - 3 = -3$$

$$2x = 0$$

$$x = 0$$

17) $\lim_{x \rightarrow 0} \frac{\sin px}{x} =$

A. $1/p$

B. 1

C. $1/2$

D. p

18) Given that $y = \tanh(2x)$, then $dy/dx =$

- for y/x
- 4-72
- A. $\text{sech}^2 2x$
 - B. $2\text{sech}^2 2x$
 - C. $\frac{1}{2}\text{sech}^2 2x$
 - D. $-2\text{sech}^2 2x$

19) The probability distribution table of a discrete random variable, X is given below. The mean is

x	1	2	3
$P(X=x)$	$1/2$	$1/3$	$1/6$

- A. 1
- ~~B. $5/3$~~
- C. $1/3$
- D. $2/3$

20) The probability density function of a continuous random variable X is

$$f(x) = \begin{cases} kx(2-x), & 0 \leq x \leq 2 \\ 0, & \text{elsewhere} \end{cases}. \text{ The value of } K \text{ is}$$

- A. $3/4$
- B. $4/3$
- C. 0
- D. 1

21) The Laplace Transform pf $f(t) = 3 \sinh 2t$ is:

- A. $3/(s^2 - 4)$
- ~~B. $6/(s^2 - 4)$~~
- C. $12/(s^2 - 4)$
- D. $6/(s^2 + 4)$.

22) The Inverse Laplace Transform of $\frac{s}{s^2 + 9}$ is:

- A. $\sin 3t$
- B. $\sinh 3t$
- C. $\cosh 3t$
- D. $\cos 3t$.

23) The eigenvalues of the matrix $M = \begin{pmatrix} 1 & -4 \\ -3 & 2 \end{pmatrix}$ are:

- A. $\{-3, -1\}$
- B. $\{2, 5\}$
- C. $\{1, 3\}$
- D. $\{-2, 5\}$.

24) Which of the following is the interval of convergence of the expansion of the function,

$$f(x) = \frac{x^4}{1+3x}?$$

- A. $(-\frac{1}{3}, \frac{1}{3})$
- B. $[-\frac{1}{3}, \frac{1}{3})$

C. $\left[-\frac{1}{3}, \frac{1}{3}\right]$

D. $\left[-\frac{1}{3}, \frac{1}{3}\right]$

25) Suppose $r(x)$ is a function defined on the interval $[-4, 4]$ such that $\lim_{x \rightarrow -2} r(x) = -2$, find $\lim_{x \rightarrow -2} [r(x)]^2$.

A. 2

B. -2

C. 4

D. -4.

26) If $g(x) = \sqrt{4-x}$ is expanded as a Maclaurin's series, then what is its range of convergence?

A. $-4 < x < 4$

B. $-4 \leq x < 4$

C. $-4 < x \leq 4$

D. $-4 \leq x \leq 4$.

27) What is the equivalent of the trigonometric identity: $\operatorname{cosec}^2 x - \cot^2 x$?

A. $1 - 2 \tan^2 x$

B. $\sin^2 x - \cos^2 x$

C. $\cos^2 x - \sin^2 x$

D. 1.

28) The following information is known about a sample: $n = 100$, $\sum x = 200$, $\sum x^2 = 600$. The variance of the sample is

A. 2

B. 4

C. 5

D. 6

29) What are the first four terms of the Fibonacci sequence defined as

$c_1 = 1$, $c_2 = 4$, $c_{n+2} = c_{n+1} - 2c_n + 4$, for $n \geq 1$?

A. 1, 2, 4, 6, ...

B. 1, 4, 3, 6, ...

C. 1, 4, 6, 2, ...

D. 1, 3, 6, 7, ...

30) Given the function $f(x) = \cosh x$. Which of the following represents $f^{-1}(x)$?

A. $\ln(x + \sqrt{x^2 - 1})$

B. $\ln(x + \sqrt{x^2 + 1})$

C. $\ln(x - \sqrt{x^2 + 1})$

D. $\ln(x - \sqrt{x^2 - 1})$.

31) If $P(A \cup B) = P(A) + P(B)$, then which of the following is TRUE for the events A and B?

A. Mutually Exclusive events

B. Dependent events

C. Exhaustive events

D. Independent events

32) Which of the following is equal to the integral $I = \int 9^{3x} dx$?

A. $I = -\frac{9^{3x}}{3 \ln 9}$

B. $I = \frac{9^{3x}}{3 \ln 9}$

C. $I = \frac{9^{3x} \ln 9}{3}$

D. $I = (3 \ln 9) 9^{3x}$

33) What is the value of the finite series $\sum_{k=1}^n \left(\frac{(k-1)!}{(k+1)!} \right)$?

A. $\frac{1}{n+1}$

B. $\frac{n}{n+1}$

C. $\frac{n+1}{n}$

D. $\frac{1}{n+1} - 1$

34) Given the vectors, $a = -2i + j + 2k$, $b = 2i - 2j + k$ and $c = i - 2j + 2k$. Which of the following is true?

A. $b \cdot c = -4$

B. $a \cdot b = 8$

C. $c \cdot a = 0$

D. $c \cdot b = 4$

35) What is the value of: $\lim_{x \rightarrow 0^+} (1-x)^{-2/x}$?

A. 0

B. e^1

C. e^2

D. $+\infty$

36) If $y = \mu x^2 e^x$ is a solution to the differential equation: $y'' - 2y' + y = 2e^x$, then what is the value of μ ?

A. -2.

B. 2

C. -1

D. 1.

37) What is the domain of the function, $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by: $f(x) = \sqrt{\frac{x}{x-1}}$?

A. $\{x: 0 < x \leq 1\}$

B. $\{x: 0 \leq x < 1\}$

C. $\{x: 0 < x < 1\}$

D. $\{x: x \geq 2\}$

38) S and T are two exhaustive events with $P(S) = 0.70$ and $P(T) = 0.60$. What is the probability of the event $(A \cap B)$?

A. 0.10

B. 0.30

C. 0.42

D. 0.86

- 39) In a store, a basin contains marked down handbooks. Given that 60 % of the handbooks are spiral-bound and 70 % of the handbooks are large-squared. Among the large-squared handbooks, 20 % are spiral-bound. Larissa randomly chooses a spiral-bound handbook. What is the probability that it is large-squared?

A. 0.12

B. 0.14

C. 0.26

D. 0.54

- 40) The function $f: \mathbb{R} \rightarrow \mathbb{R}$ is defined by $f(x) = -x^3 + 2x$. Which of the following is CORRECT about $f(x)$?

A. It is periodic

B. Neither even nor odd

C. It is even

D. It is odd

SECTION - B: STRUCTURAL

(60 MARKS)

Part - 1: Analysis (30 Marks)

- 1) Given the function $f(x) = \frac{4\sqrt{x} - 4}{x^2 - 1}$

a). State the domain of definition if f

b). Evaluate the limit of f as x tends to $+1$

- 2) Evaluate $\int_0^1 \frac{\sqrt{x^2 - 9}}{x} dx$

- 3) a. Find the inverse Laplacean of $L^{-1} \left\{ \frac{s+5}{s^2 - 2s - 3} \right\}$

b. By using Laplace Transform, solve the differential equation:

$$y'' + 4y = e^t, y(0) = 2, y'(0) = 1$$

- 4) Find the Fourier coefficients of the periodic function

$$F(x) = \begin{cases} 0, & -\pi < x < 0 \\ x, & 0 < x < \pi \end{cases}$$

Part - 2: Statistics (30 Marks)

- 2.1 The density of molten salt mixtures y , in gcm^{-3} , was measured at various temperatures x °C. The results were:

Temperature (x)	250	270	290	310	330	350
Rank x	7	8	9	10	11	12
Density (y)	1.955	1.935	1.890	1.920	1.895	1.865
Rank y	6	5	2	4	3	1

$\sum R_x = 57$

$\sum R_y = 21$

- Plot a graph of y against x showing these points (the graph is called a scatter diagram) (4 marks)
- Determine the equation of the regression line of y on x . (6 marks)
- Determine the equation of the regression line of x on y . (2 marks)
- Plot the regression lines in (b) and (c) on the same graph as the scatter diagram in (a). (4 marks)
- Calculate the product moment correlation coefficient for the distribution. (6 marks)
- Calculate the Spearman's correlation coefficient of rank. (4 marks)
- Calculate the Kendall correlation coefficient of ranks (4 marks)

PART - 3: PROBABILITY (20 marks)

3.1 A bag contains 7 green small spheres, 14 red small spheres and 29 yellow small spheres, all of which are identical to touch. Three small spheres are drawn at random from the bag, one after the other. Determine the probabilities of;

- Getting one sphere of each colour with replacement, (2 marks)
- Getting one sphere of each colour without replacement, (2 marks)
- Getting either one green small sphere and two red small spheres or one yellow and two red small spheres. (5 marks)

3.2 Given that 10 percent of candidates sitting for an HND examination are left-handed, use Poisson distribution to estimate the probability that a random sample of 100 candidates contains 2 or more left-handed candidates. (4 marks)

3.3 A large number of candidates sat for an ENGLISH Examination. The marks obtained can be regarded as a continuous variable which is normally distributed. Given that 14 % of the candidates obtained less than 30 marks, and $24\frac{1}{2}\%$ obtained more than 50 marks, find the mean and variance of the distribution, giving your answer to one decimal place. (7 marks)